

Question ID: 114ae8a9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

A train traveled **173** miles in the first **3** hours of a trip and traveled at an average speed of **53** miles per hour for the remainder of the trip. Which equation gives the total number of miles, *y*, the train traveled if the trip was *x* hours long and *x* > 3?

Answer

- A. *y* = 53*x* + 173
- B. *y* = 53*x* + 14
- C. *y* = 14*x* + 226
- D. *y* = 14*x* + 53

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question
For the linear function h , $h(0) = 0$ and $h(11) = 10$. Which equation defines h ?

- Answer**
- A. $h(x) = \frac{10}{11}x$
 - B. $h(x) = \frac{10}{11}x + 10$
 - C. $h(x) = \frac{11}{10}x$
 - D. $h(x) = \frac{11}{10}x + 11$

Question ID: 74d0906a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

A tow truck travels at a speed of at least **25** miles per hour but no more than **45** miles per hour for a section of a road. Which inequality represents this situation, where x is the speed of the tow truck, in miles per hour, on this section of the road?

Answer

- A. $x \geq 25$
- B. $x \geq 45$
- C. $25 \leq x \leq 45$
- D. $x \leq 70$

Question ID: 8383771a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$x + y = 20$

One of the two equations in a system of linear equations is given. The system has exactly one solution. Which equation could be the second equation in the system?

Answer

- A. $3x + 3y = 60$
- B. $-x - y = 40$
- C. $3x + 3y = 14$
- D. $x - y = 20$

Question ID: 8b9960b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

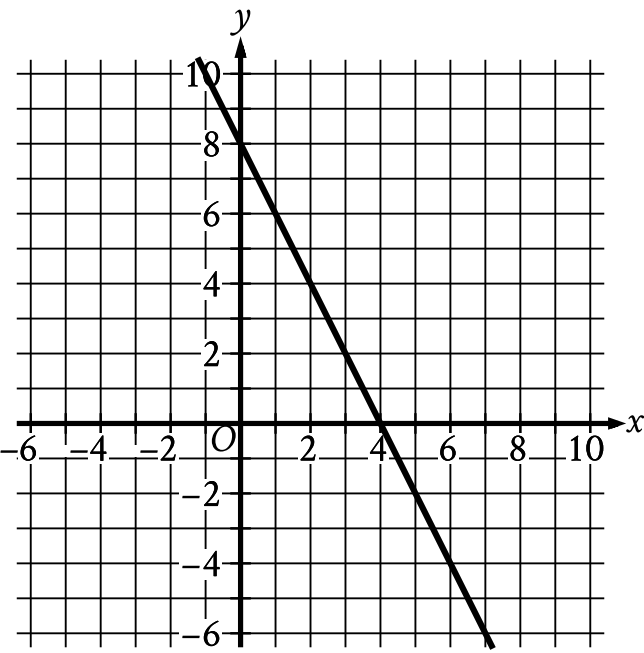
The owner of a bakery is giving away free muffins every hour. The equation $f(x) = 108 - 9x$ gives the number of free muffins, $f(x)$, that remain to be given away after x hours. Which of the following is the best interpretation of $f(12) = 0$ in this context?

Answer

- A. The owner of the bakery has 12 free muffins to give away.
- B. The owner of the bakery will give away 12 free muffins in 9 hours.
- C. It will take 12 hours to give away all the free muffins.
- D. It will take 108 hours to give away 12 free muffins.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

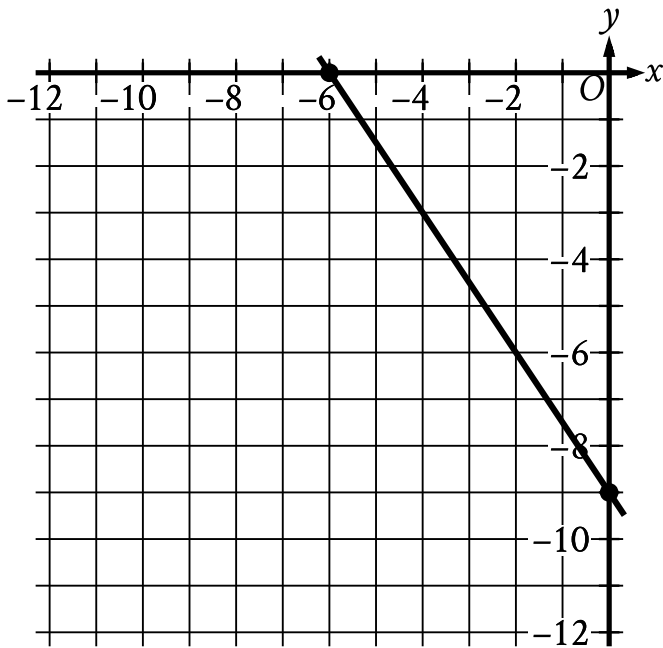


The graph of the linear function f is shown, where $y = f(x)$. Which equation defines function f ?

- Answer
- A. $f(x) = -2x - 8$
 - B. $f(x) = -2x + 8$
 - C. $f(x) = 2x - 8$
 - D. $f(x) = 2x + 8$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question



The graph of line h is shown in the xy -plane. Line k (not shown) is defined by $sx + 48y = t$, where s and t are constants. If line k is graphed in this xy -plane, the result is a graph of two linear equations. This system of two linear equations has no solution. Which of the following is **NOT** a possible value of t ?

Answer

- A. **—432**
- B. **—9**
- C. **72**
- D. **288**

Choice C is incorrect. This is the value of s , not a value of t that isn't possible.

Choice D is incorrect. If $t = 288$, then the y -intercept of line k is $(0, \frac{288}{48})$, or $(0, 6)$, which is different from the y -intercept of line h , $(0, -9)$. Therefore, **288** is a possible value of t .

Question ID: c039b198

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

A computer company pays customers for trading in old computers. The payment p , in dollars, is the original retail price of a computer minus \$150 for each year since the computer was released. Which equation represents this situation for a computer with an original retail price of \$690 when it was released t years ago, where t is a whole number and $1 \leq t \leq 4$?

Answer

- A. $p = 150 - 690t$
- B. $p = 690 - 150t$
- C. $t = 150 - 690p$
- D. $t = 690 - 150p$

Question ID: c387583f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

In the xy -plane, line k passes through the points $(8, 6)$ and $(9, 12)$ and is defined by the equation $y = mx + b$. What is the value of b ?

Question ID: e142f83c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$y - 6 = \frac{4}{5}(x + 5)$$

$$y - v = \frac{4}{5}(x + 10)$$

In the given system of equations, v is a constant. The system has infinitely many solutions. What is the value of v ?

Answer

- A. 0
- B. 2
- C. 4
- D. 12

Question ID: ee1f63c0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function f is defined by $f(x) = 6.2x$. What is the value of $f(x)$ when $x = 10$?

Answer

- A. 72
- B. 62
- C. 17.2
- D. 16.2

Question ID: 0de8359f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$r(t) = 63t - 2t^2$

The function r is defined by the given equation. The function s is defined by $s(t) = r(t) + 1$. Which expression represents the maximum value of $s(t)$?

Answer

- A. $1 + \left(\frac{63}{2}\right)^2$
- B. $1 + 2\left(\frac{63}{4}\right)^2$
- C. $1 - 2\left(\frac{63}{4}\right)^2$
- D. $1 - \left(\frac{63}{2}\right)^2$

Question ID: 405c5ced

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

The function C defined by $C(t) = 6(1.5)^t$ gives the estimated number of clients at a certain business, where t is the number of weeks since the business opened. Which of the following is the best interpretation of the number **6** in this context?

Answer

- A. The estimated number of clients when the business opened
- B. The increase in the estimated number of clients each week
- C. The number of weeks the business has been open
- D. The percent increase in the estimated number of clients each week

Question ID: 512bd5b4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = 27(1.20)^{\frac{x}{3}}$

For the given function f , the value of $f(x)$ increases by $p\%$ for every increase of x by 6. What is the value of p ?

Question ID: 53c07bf9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

The value of a certain fishing boat was **\$24,000** when it was purchased. Each year, the value of the fishing boat decreased by **20%** of the previous year’s value. Which equation represents this situation, where *v* is the value of the fishing boat, in dollars, and *y* is the number of years since the fishing boat was purchased?

Answer

- A. $v = 0.8(24,000)^y$
- B. $v = 1.2(24,000)^y$
- C. $v = 24,000(0.8)^y$
- D. $v = 24,000(1.2)^y$

Question ID: 5ba187d4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $100x^2 + 2,600$?

Answer

- A. $100(26x^2)$
- B. $100(x^2 + 26)$
- C. $2,600(100x^2)$
- D. $2,600(100x^2 + 1)$

Question ID: 9833d0e3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

According to a model, the estimated volume of a certain tree in a forest increases by **13.9%** for every increase of **100** centimeters in the tree's height. The equation $y = ab^{\frac{x}{100}}$ represents this model, where y represents the estimated volume, in cubic centimeters, of this tree in the forest, x represents the tree's height, in centimeters, and a and b are constants. In this equation, what is the value of b ?

Answer

- A. **0.139**
- B. **1**
- C. **1.139**
- D. **13.9**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which expression is equivalent to $\frac{4x+1}{x-9}$, where $x > 9$?

Answer

- A. $4 + \frac{37}{x-9}$
- B. $4 + \frac{35}{x-9}$
- C. $4 + \frac{1}{x-9}$
- D. $4 - \frac{1}{x-9}$

Question ID: b5ebd31c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$2r = 9s + t$

The given equation relates the variables r , s , and t . Which equation correctly expresses s in terms of r and t ?

Answer

- A. $s = \frac{2}{9}r - t$
- B. $s = \frac{2r-t}{9}$
- C. $s = 18r - t$
- D. $s = 9(2r - t)$

Question ID: befeb95f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The function f is defined by $f(x) = x^3 + 15$. What is the value of $f(2)$?

Question ID: c5d83a99

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$\frac{1}{x} + 8 = \frac{1}{6} + 8$

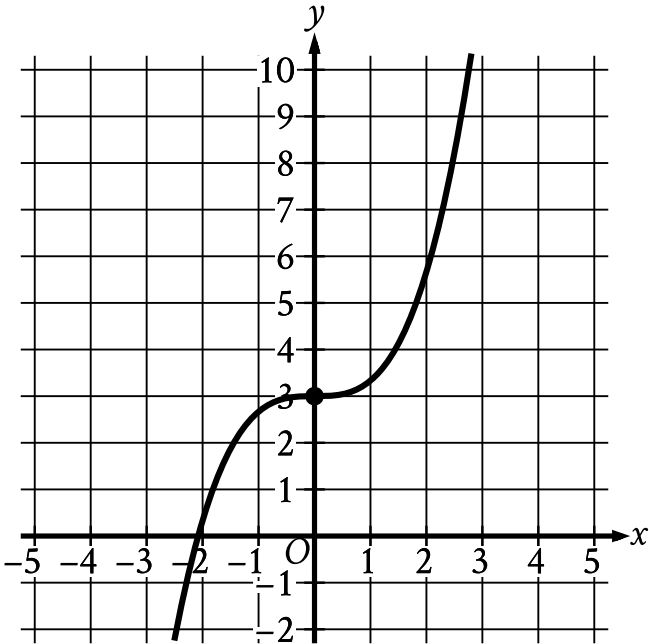
What is the solution to the given equation?

Answer

- A. 1
- B. 6
- C. 8
- D. 14

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question



The graph of $y = f(x)$ is shown. What is the y-intercept of $y = f(x) + 5$?

Answer

- A. $(0, 8)$
- B. $(0, 5)$
- C. $(0, 3)$
- D. $(0, 0)$

Question ID: de91a6fc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A certain bowl was sold online for prices varying between **\$7** and **\$13**. The number of bowls sold can be modeled by a quadratic function based on the price, where **5** bowls were sold for **\$12** each and the maximum of **9** bowls were sold for **\$10** each. According to the model, what is the number of bowls that were sold for **\$8** each?

Question ID: 0d125b35

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Medium

Question

There is a proportional relationship between p and t . If $p = 5,130$ when $t = 6,840$, then what is the value of p when $t = 5,472$?

Answer

- A. 7,296
- B. 7,182
- C. 4,104
- D. 3,762

Question ID: 1f4c57d7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

For the last three years, a pig at a farm gave birth to one litter per year. The table shows the number of piglets in each of these three litters.

Litter number	Number of piglets
1	8
2	7
3	9

What is the mean number of piglets per litter for these three litters?

Answer

- A. 2
- B. 7
- C. 8
- D. 9

Question ID: 3aa0642c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

Shape	Number of cards
Square	5
Circle	9
Triangle	14
Star	12

The table summarizes the distribution of shape printed on one side for cards in a deck of cards. How many cards in the deck have a triangle printed on one side?

Question ID: 589cd7a6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Easy

Question

A box holds **104** tokens, of which **13** are brown. If a token is selected at random from the box, what is the probability of selecting a token that is brown?

Answer

- A. $\frac{1}{13}$
- B. $\frac{1}{10}$
- C. $\frac{1}{8}$
- D. $\frac{1}{4}$

Question ID: 81d0f1e6

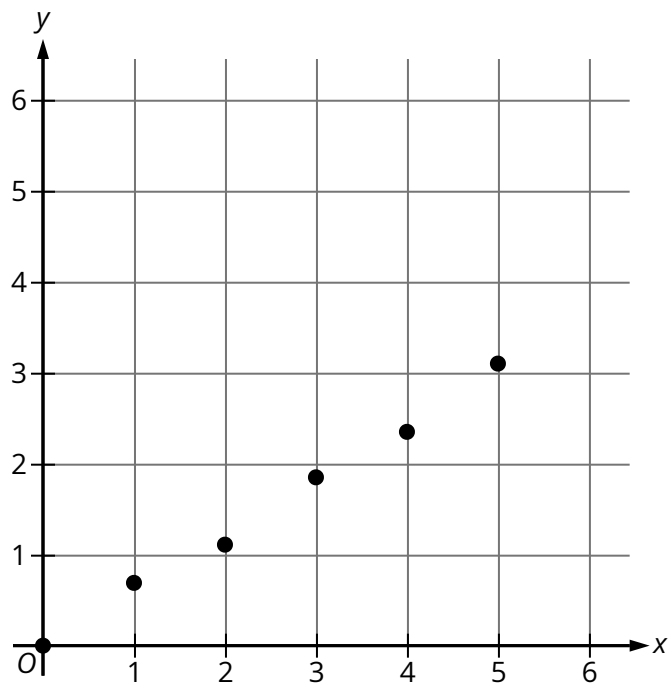
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Medium

Question

The ratio a to b is equivalent to the ratio 63 to 17 . What is the value of a when $b = 153$?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Easy

Question



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

Answer

- A. $y = -0.606x$
- B. $y = 0.606 - 1.001x$
- C. $y = 0.606x$
- D. $y = 0.606 + 1.001x$

Choice A is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice B is incorrect. The graph of this model has a negative slope, not a positive slope, and a y -intercept that is not $(0, 0)$.

Choice D is incorrect. The graph of this model has a slope that is too steep to fit the data, and a y -intercept that is not $(0, 0)$.

Question ID: a51e6f93

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

0.1152 is 0.08% of the number b . The number b is 25% of the number c . What is the value of c ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard

Question

A statistician conducted a study to investigate the relationship between the average number of minutes piano students practiced each week and the percentage of correct notes the students played during a recital. For the study, the statistician selected a sample of **85** piano students at random from all the piano students in a certain region who were in their second year with their conservatory. The table shows the information for **75** piano students in this sample who practiced at least **50** minutes on average each week.

	Percentage of correct notes the students played during a recital			
Average number of minutes practiced each week	Less than 50%	50% to 80%	Greater than 80%	Total
50 to 150	7	28	5	40
151 or more	3	11	21	35
Total	10	39	26	75

Which of the following is the largest population to which the results of the study can be generalized?

Answer

- A. All piano students who were in their second year with their conservatory
- B. All piano students in the region
- C. All piano students in the region who were in their second year with their conservatory
- D. The **75** piano students in the sample who practiced at least **50** minutes on average each week

Question ID: b73025e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Easy

Question

How many cups are equivalent to **83** quarts? (**4 cups = 1 quart**)

Answer

- A. **21**
- B. **79**
- C. **87**
- D. **332**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Data Set Q

Value	Frequency
a	110
b	30
c	0

Data Set R

Value	Frequency
a	0
b	3
c	11

Data sets Q and R are summarized in the tables shown, where a , b , and c are positive, consecutive integers and $a < b < c$. Which statement comparing the means of the data sets is true?

Answer

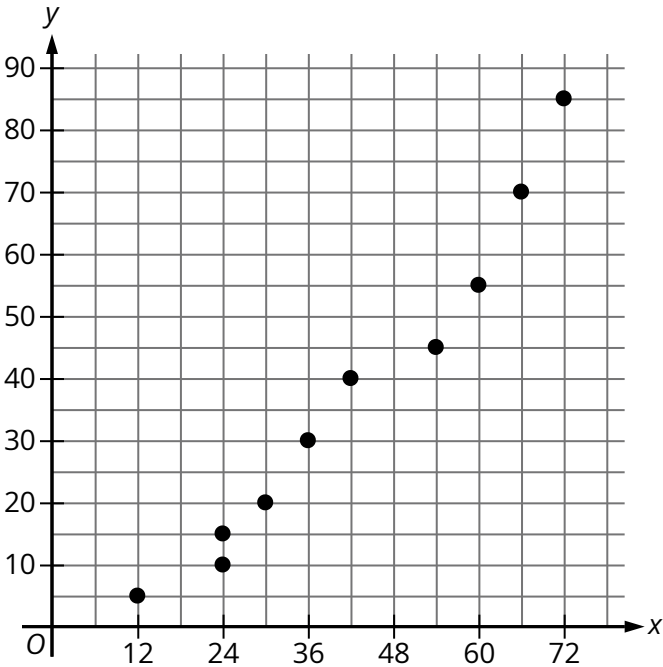
- A. The mean of data set R is $\frac{2(11)}{14}$ less than the mean of data set Q.
- B. The mean of data set R is $\frac{2(11)}{14}$ greater than the mean of data set Q.
- C. The mean of data set Q is **10** times the mean of data set R.
- D. The means of the two data sets are equal.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Medium

Question

The owner of a restaurant supply store selects **10** cutting boards to put on sale. The graph shows the width x , in centimeters, of each cutting board and its corresponding sale price y , in dollars, where each value of y is an integer.



According to the graph, which statement is true?

- Answer**
- A. Cutting boards that have a width greater than **48** centimeters have an average sale price of **\$59.00**.
 - B. Cutting boards that have a width greater than **48** centimeters have an average sale price of **\$40.00**.
 - C. Cutting boards that have a width between **18** and **48** centimeters have an average sale price of **\$31.20**.
 - D. Cutting boards that have a width between **18** and **48** centimeters have an average sale price of **\$23.00**.

Choice C is incorrect. Cutting boards that have a width between **18** and **48** centimeters have an average sale price of **\$23.00**, not **\$31.20**.

Question ID: ed6b7e5f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Hard

Question

The table gives the distribution of topping and type of crust for customer orders at a pizza place.

Topping	Type of crust	
	Thin	Thick
Extra cheese	60	30
Pepperoni	20	30
Tomato	80	20

If one of these customer orders is selected at random, what is the probability of selecting an order with thin crust, given the topping is pepperoni?
(Express your answer as a decimal or fraction, not as a percent.)

Question ID: f85bbf7a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Medium

Question

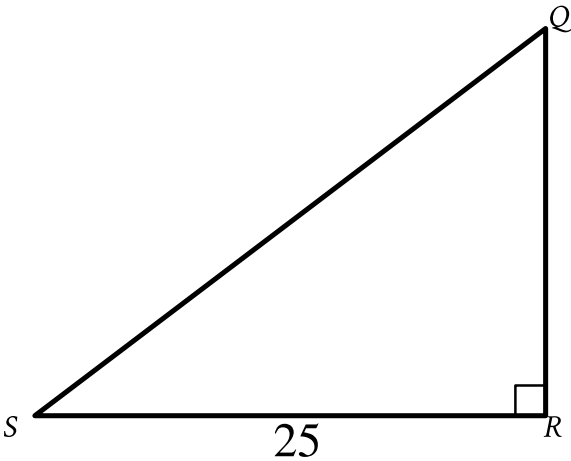
A middle school faculty surveyed students to determine interest in an after-school chess club. Of the **100** students who were selected at random from all students in the school, **32%** expressed interest in the club. Based on the sample mean and the margin of error, the faculty expects that between **24%** and **40%** of all students in the school would express interest in the club. What is the margin of error?

Answer

- A. 0%
- B. 8%
- C. 16%
- D. 32%

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

- Answer
- A. $25 \cos Q$
 - B. $25 \sin Q$
 - C. $\frac{25}{\cos Q}$
 - D. $\frac{25}{\sin Q}$

Question ID: 27d075d8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question
For two acute angles, $\angle Q$ and $\angle R$, $\cos (Q)=\sin (R)$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x+52$ and $4x+13$, respectively. What is the value of x ?

Answer

- A. 5
- B. 13
- C. 23
- D. 38

Question ID: 2a549721

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question
Rectangle $ABCD$ has an area of **280** square centimeters. The length of the rectangle is **10** centimeters. What is the width, in centimeters, of the rectangle?

- Answer**
- A. **10**
 - B. **28**
 - C. **76**
 - D. **140**

Question ID: 32f99025

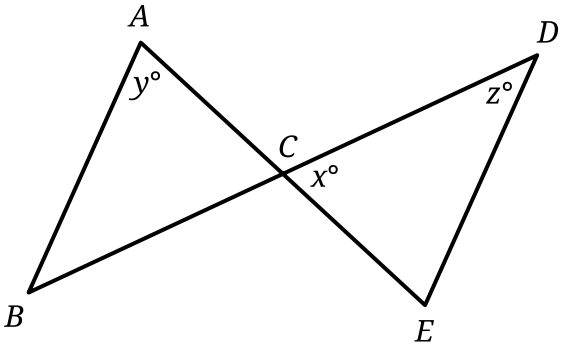
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question
The length of the base of a triangle is **8** inches, and the height of the triangle is **12** inches. What is the area, in square inches, of the triangle?

- Answer**
- A. **20**
 - B. **48**
 - C. **60**
 - D. **96**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure, $\overline{AE}$ intersects $\overline{DB}$ at point C and $\overline{AB}$ is parallel to $\overline{DE}$, creating triangles ABC and CDE . If $x = 63$ and $y = 2x - 47$, what is the value of z ?

Answer

- A. 38
- B. 47
- C. 63
- D. 79

Question ID: 4ddfa209

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

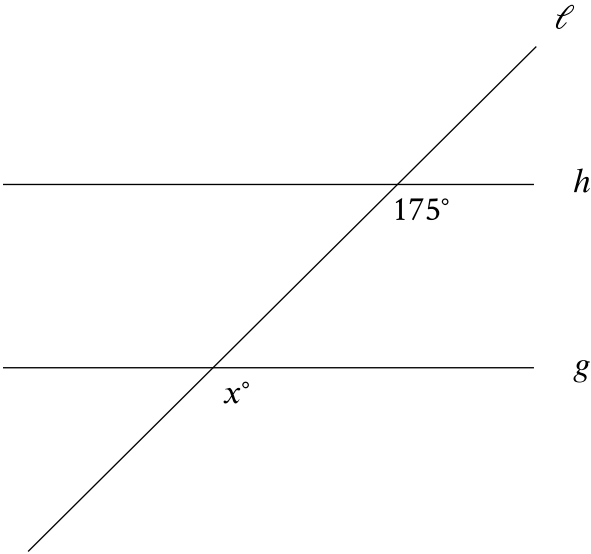
The length of one side of an equilateral triangle is **28** inches. What is the perimeter, in inches, of this triangle?

Answer

- A. **28**
- B. **31**
- C. **84**
- D. **112**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure shown, line g is parallel to line h . Line **Math output error** intersects both lines. What is the value of x ?

Question ID: 62d39eb8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

What is the radius of the circle in the xy -plane defined by $(x + 3)^2 + (y + 5)^2 = 289$?

Question ID: c0b53183

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

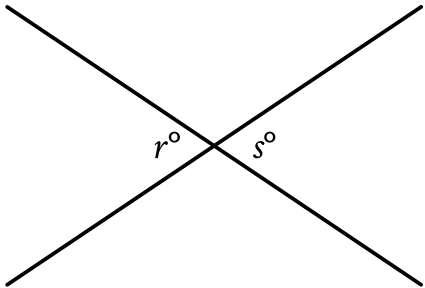
Question

A circle has center O , and points A and U lie on the circle. Line segments AE and UE are tangent to the circle at points A and U , respectively. The distance between point O and point E is 510 centimeters and the distance between point O and point U is 78 centimeters. What is the area, in square centimeters, of quadrilateral $OAEU$?

Question ID: d71550d0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $r = 53$, what is the value of s ?

- Answer
- A. 37
 - B. 53
 - C. 143
 - D. 233

Question ID: e570a99c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question
In $\triangle JKL$, the measures of both $\angle J$ and $\angle K$ are equal and the measure of $\angle L$ is 118° . What is the measure of $\angle J$?

- Answer**
- A. 31°
 - B. 59°
 - C. 62°
 - D. 90°

Question ID: ebf3a962

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

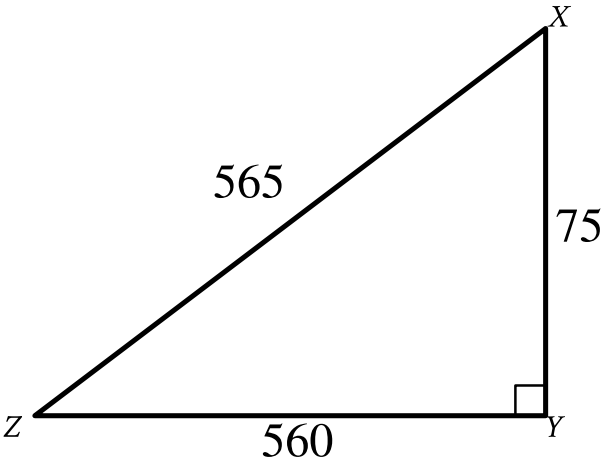
Question

A right circular cylinder has a height of **16** inches and a volume of **$3,136\pi$** cubic inches. What is the radius, in inches, of this cylinder?

Question ID: f1251f35

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

In triangle XYZ , $\cos Z = \frac{560}{c}$. What is the value of c ?